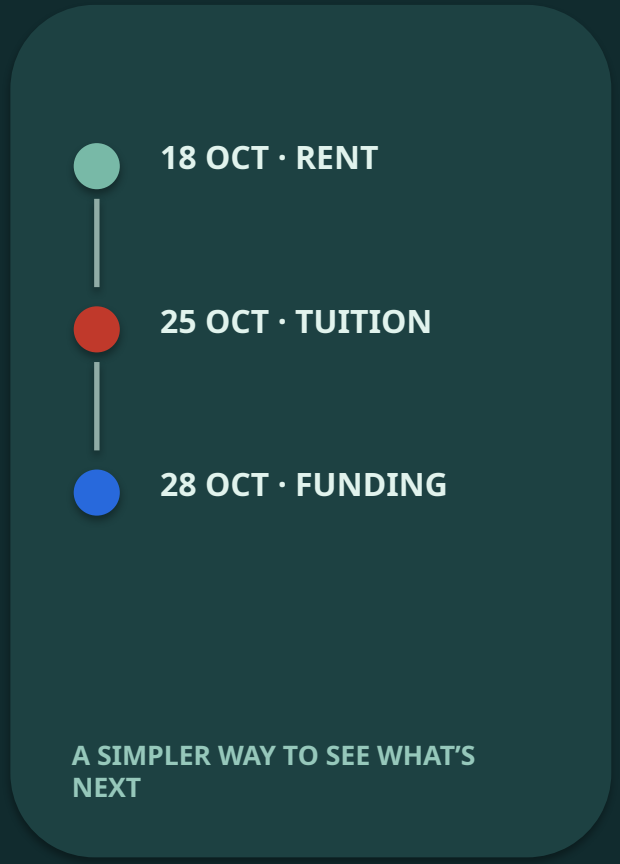
TOPIC C · INTERNATIONAL STUDENT FINANCE

Their money has a timeline.

CAMPUSFIN AI / PREDICT · RESCUE · PROVE

Deadline-aware financial recovery for international students



Financial stress is often a timing problem.

Students receive uncertain funding. Essential obligations still have fixed deadlines.

01

Cross-border delay

An expected family transfer may arrive after fees become due.

02

Fragmented information

Remittances, scholarships and wages sit outside the same forecast.

03

Unverified advice

A plausible AI suggestion may be infeasible without confirmed money.

THE GOAL · Help a student act before timing becomes a missed tuition payment.

Meet Amina. One delayed transfer changes everything.

A reproducible hypothetical case set in Kuala Lumpur. No real bank records.

OPENING CASH

RM 1,200

Includes previously received wages

RENT · 18 OCT

RM 700

Fixed obligation

TUITION · 25 OCT

RM 1,500

Fixed obligation

Family remittance · USD 350

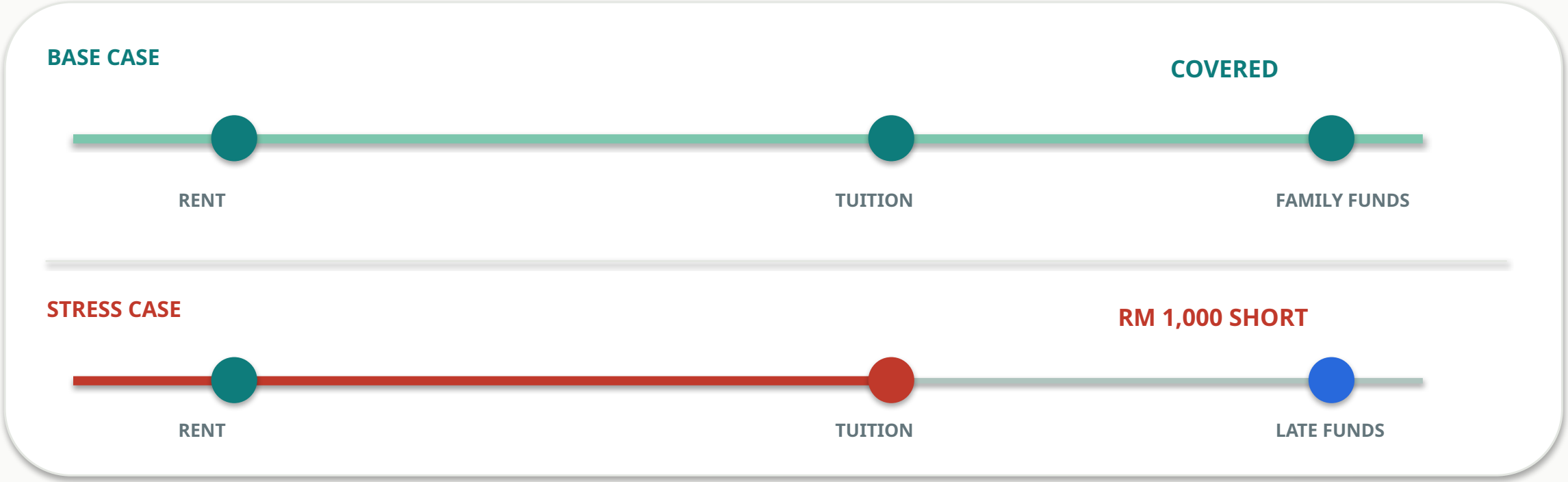
USD 5 deducted fee × 4.20 MYR/USD = RM 1,449 net

DELAYED TO 28 OCTOBER

RM 1,000 GAP

One change. A visible financial collision.

The Financial Horizon recomputes projected funding as soon as a scenario changes.



A responsible rescue can also say “not feasible.”

Conditional actions are simulated before they are recommended.

EARLIER TRANSFER

CONDITIONAL

Requires sender funds available and timely arrival.

UNIVERSITY EXTENSION

PENDING APPROVAL

Never treated as accepted until confirmed.

FAMILY TOP-UP

CONDITIONAL

Extra amount sized using verified math.

ESSENTIALGUARD

A tuition solution should not leave the student without essentials.

Decision X-Ray: show why, not just what.

Inspect the exact change, the source records, and the remaining assumptions.

INPUT

USD 350 family transfer

→

CHANGE

18 Oct → 28 Oct

→

CONSTRAINT

Tuition due 25 Oct

→

RESULT

RM 1,000 gap

PROOFCARD **Every displayed amount can be traced to ledger inputs and the Python engine.**

One supervised agent. Deterministic money math.

Gemini explains and orchestrates read-only tools; the backend calculates all financial values.

STUDENT QUESTION

“What if my transfer arrives late?”

→

GEMINI COPILOT

Approved tool calls
No ledger writes

→

PYTHON ENGINE

Forecast, shortfalls
Verified fact cards

SECURITY RULE **No AI permission to settle funds, send money, or approve a request.**

An Innovation Lab judges can challenge.

Five deliberately connected capabilities—not twenty decorative features.

01 **CRISIS TIME MACHINE**

Four recalculated futures

02 **ESSENTIALGUARD**

Protect student reserves

03 **DECISION X-RAY**

Auditable reason codes

04 **FX SHOCK ABSORBER**

Hypothetical currency swings

05 **CHAOS LAB**

Seven fixture-based checks

THE CREATIVE CORE

A visible “why” behind each choice.

Trust is a behavior we test.

Automated Python tests are recorded; unexecuted QA is disclosed honestly.

72 / 72

BACKEND TESTS PASSED

DETERMINISTIC FINANCE

Decimal amount calculations

UNAUTHORIZED ACTIONS

Read-only Gemini tool allowlist

SCENARIO CONSISTENCY

Chat, Horizon and ProofCard align

CHAOS LAB

Actual fixture assertions, not fake output

Frontend production build & device QA: final Windows verification required.

Built to demonstrate financial agency safely.

React + FastAPI + SQLite + Gemini, with a verified calculation boundary.

REACT

Mobile-first premium UI

FASTAPI

Validation + permissions

DECIMAL ENGINE

Forecast + rescue math

SQLITE

Local demo + audit

CLEARLY OUT OF SCOPE

Live banking, real money transfers, authenticated public access, official FX quotes, university approvals and production compliance.

Predict the crisis.
Prove the options.
Protect the student.

CAMPUSFIN AI

PREDICT · RESCUE · PROVE